

Exercise Set 2

Problem 1

a) Which of the expressions below correspond to the statement: “the probability of rain on Monday?”

1. $\Pr(\text{rain})$
2. $\Pr(\text{rain}|\text{Monday})$
3. $\Pr(\text{Monday}|\text{rain})$
4. $\Pr(\text{rain}, \text{Monday}) / \Pr(\text{Monday})$

Correct Answer: Expression 2 and 4 state "the probability of rain on Monday"

Interpretation:

1. Probability of rain.
2. Probability of rain, given that it is Monday.
3. Probability of it being Monday, given rain.
4. Probability of rain and it being Monday, given that it is Monday, as:

The joint probability is defined as: $\Pr(w, p) = \Pr(w|p) \Pr(p)$

Written in terms of rain and Monday: $\Pr(\text{rain}, \text{Monday}) = \Pr(\text{rain}|\text{Monday}) \Pr(\text{Monday})$ or:

$$\frac{\Pr(\text{rain}, \text{Monday})}{\Pr(\text{Monday})} = \Pr(\text{rain}|\text{Monday})$$

b) Convert each statement into a probability expression. Which of the following statements corresponds to the expression: $\Pr(\text{Monday}|\text{rain})$?

1. The probability of rain on Monday.
2. The probability of rain, given that it is Monday.
3. The probability that it is Monday, given that it is raining.
4. The probability that it is Monday and that it is raining.

1. $\Pr(\text{rain}|\text{Monday})$
2. $\Pr(\text{rain}, \text{Monday})$
3. $\Pr(\text{Monday}|\text{rain})$
4. $\Pr(\text{Monday}, \text{rain})$

Correct answer: Expression 3

c) Which of the expressions below correspond to the statement: “the probability that it is Monday, given that it is raining?”

1. $\Pr(\text{Monday}|\text{rain})$
2. $\Pr(\text{rain}|\text{Monday})$
3. $\Pr(\text{rain}|\text{Monday}) \Pr(\text{Monday})$
4. $\Pr(\text{rain}|\text{Monday}) \Pr(\text{Monday}) / \Pr(\text{rain})$
5. $\Pr(\text{Monday}|\text{rain}) \Pr(\text{rain}) / \Pr(\text{Monday})$

Interpretation:

- **Expression 1:** probability that it is Monday, given that it is raining.
- **Expression 2:** probability of rain, given that it is Monday.
- **Expression 3:** $\Pr(\text{rain}|\text{Monday})\Pr(\text{Monday}) = \Pr(\text{rain}, \text{Monday})$, thus: the probability that it is raining and that it is Monday.
- **Expression 4:** can be rewritten as $\Pr(\text{rain}, \text{Monday})/\Pr(\text{rain})$ and simplified using the joint probability formula to $\Pr(\text{Monday}|\text{rain})$. This is the probability that it is Monday, given that it is raining.
- **Expression 5:** Same as expression 4 but with the terms exchanged. Repeating the previous steps, it is the probability of rain, given that it is Monday.

Correct answers: Expression 1 and 4.

Problem 2

Suppose there are two globes, one for Earth and one for Mars:

- Earth is 70% covered in water.
- Mars is 100% land.

Further suppose that one of these globes—you don't know which—was tossed in the air and produces a "land" observation. Assume that each globe was equally likely to be tossed. Show that the posterior probability that the globe was the Earth, conditional on seeing "land" ($\Pr(\text{Earth}|\text{land})$), is 0.23.

$$\Pr(\text{land}|\text{Earth}) = 1 - 0.7 = 0.3$$

$$\Pr(\text{land}|\text{Mars}) = 1$$

$$\Pr(\text{Earth}) = \Pr(\text{Mars}) = 0.5$$

Now, we need to use Bayes' theorem to get the answer:

$$\Pr(\text{Earth}|\text{land}) = \frac{\Pr(\text{land}|\text{Earth})\Pr(\text{Earth})}{\Pr(\text{land})} = \frac{0.3(0.5)}{\Pr(\text{land})} = \frac{0.15}{\Pr(\text{land})}$$

We still need to calculate $\Pr(\text{land})$. This is called the marginal likelihood, and to calculate it, we need to take the probability of each possible globe and multiply it by the conditional probability of seeing land given that globe; we then add up every such product:

$$\Pr(\text{land}) = \Pr(\text{land}|\text{Earth})\Pr(\text{Earth}) + \Pr(\text{land}|\text{Mars})\Pr(\text{Mars}) = 0.3(0.5) + 1(0.5) = 0.65$$

Now we can substitute this value into the formula from before to get our answer:

$$\Pr(\text{Earth}|\text{land}) = \frac{0.15}{\Pr(\text{land})} = \frac{0.15}{0.65} = 0.23$$

References

McElreath, R. (2020). *Statistical rethinking- A Bayesian course with examples in R and Stan*. CRC press